# Alert Triage Report

## Introduction

This practical exercise focused on alert triage and automated validation of security events using SIEM and threat intelligence tools. The objective was to analyze alerts generated in Wazuh, prioritize them based on severity, and automate validation using VirusTotal integrated with TheHive.

The exercise demonstrates how Security Operations Centers (SOCs) can efficiently handle large volumes of alerts while reducing manual effort through automation.

## Tools Used

| Tool | Purpose |
|---|---|
| Wazuh | Alert generation and log monitoring |
| TheHive | Incident response and case management |
| VirusTotal | Threat intelligence and file/hash analysis |

## Alert Overview

A simulated alert was generated in Wazuh indicating a suspicious file download activity.
Such alerts may indicate malware delivery or unauthorized access attempts.

## Alert Triage Process

The triage process involved the following steps:

1. **Alert Identification**
    - Review alert details in Wazuh dashboard.
2. **Initial Analysis**
    - Check source IP and file activity.
    - Determine if behavior is suspicious.
3. **Prioritization**
    - Assign severity based on risk level.
4. **Case Creation**
    - Forward alert to TheHive for investigation.

### Triage Data

| Alert ID | Description | Source IP | Priority | Status |
|---|---|---|---|---|
| 005 | File Download | 192.168.1.102 | High | Open |

### Observation

The alert was marked as **High priority** due to the possibility of malicious file execution and potential compromise of the system.

## Automated Validation Workflow

To enhance triage efficiency, automated validation was configured using TheHive and VirusTotal.

### Workflow Steps

1. Alert forwarded from Wazuh to TheHive
2. Extract file hash from alert
3. Automatically query VirusTotal API
4. Retrieve reputation score and detection results
5. Attach results to TheHive case

## Validation Results

- The file hash was flagged as **malicious** by multiple antivirus engines
- VirusTotal provided threat classification and detection ratio
- TheHive case was automatically enriched with intelligence data

This automation significantly reduced manual investigation time.

## Threat Analysis

The alert indicates a potential malware delivery attempt through file download. The malicious classification of the file confirms a high-risk threat that could lead to:

- System compromise
- Data exfiltration
- Persistence mechanisms

Immediate response actions such as isolation and blocking are recommended.

# Automation Benefits

- Faster alert validation
- Reduced analyst workload
- Improved accuracy in threat detection
- Centralized incident management
- Scalable SOC operations

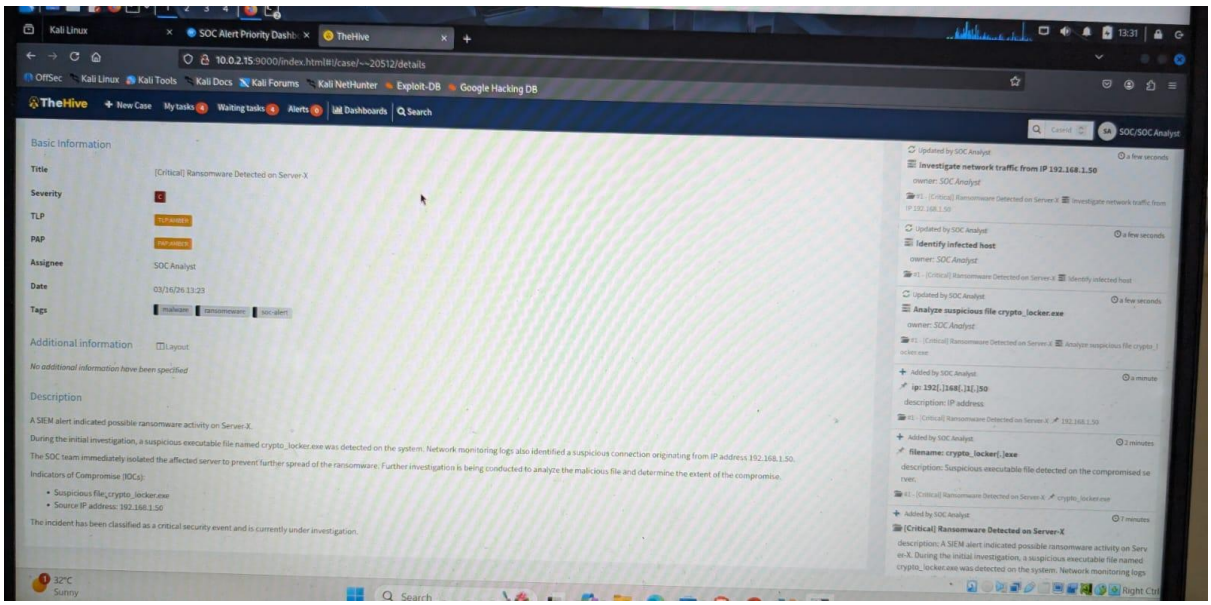
# Validation Summary

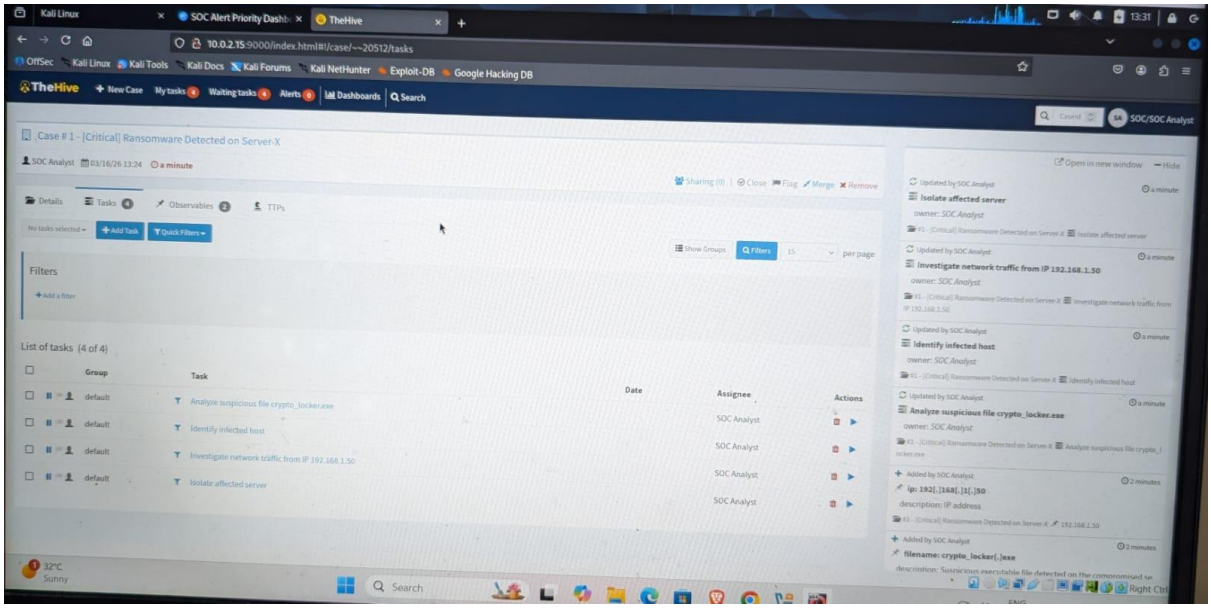
The automated validation process used TheHive to query VirusTotal for file hash analysis. The results confirmed the file as malicious based on multiple detections. This automation improved triage efficiency by quickly enriching alerts with threat intelligence, enabling faster decision-making and reducing manual effort in investigating suspicious file downloads.

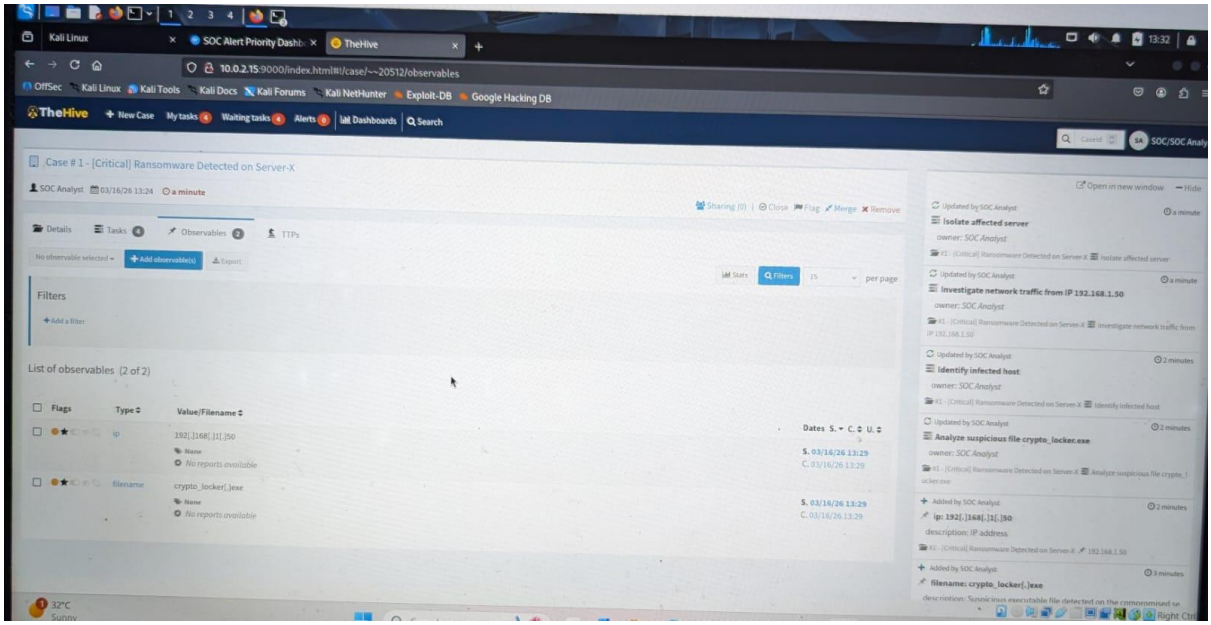
# Conclusion

This practical exercise demonstrated the importance of efficient alert triage combined with automated threat intelligence validation. By integrating Wazuh, TheHive, and VirusTotal, the SOC workflow was enhanced to quickly identify and respond to potential threats.

Automation plays a critical role in modern security operations by improving response time, accuracy, and scalability in handling security alerts.




Kali Linux
SOC Alert Priority Dashb
TheHive
10.0.2.15:9000/index.html#!/case/~-20512/tasks
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB
TheHive New Case My tasks Waiting tasks Alerts Dashboards Search
Case # 1 - [Critical] Ransomware Detected on Server-X
SOC Analyst 03/16/26 13:24 a minute
Details Tasks Observables TTPs
Filters
List of tasks (4 of 4)
Group Task Date Assignee Actions
default Analyze suspicious file crypto_locker.exe SOC Analyst
default Identify infected host SOC Analyst
default Investigate network traffic from IP 192.168.1.50 SOC Analyst
default Isolate affected server SOC Analyst
Isolate affected server
Investigate network traffic from IP 192.168.1.50
Identify infected host
Analyze suspicious file crypto_locker.exe
ip: 192[.]168[.]1[.]50
description: IP address
filename: crypto_locker[.]exe


Kali Linux
SOC Alert Priority Dashb
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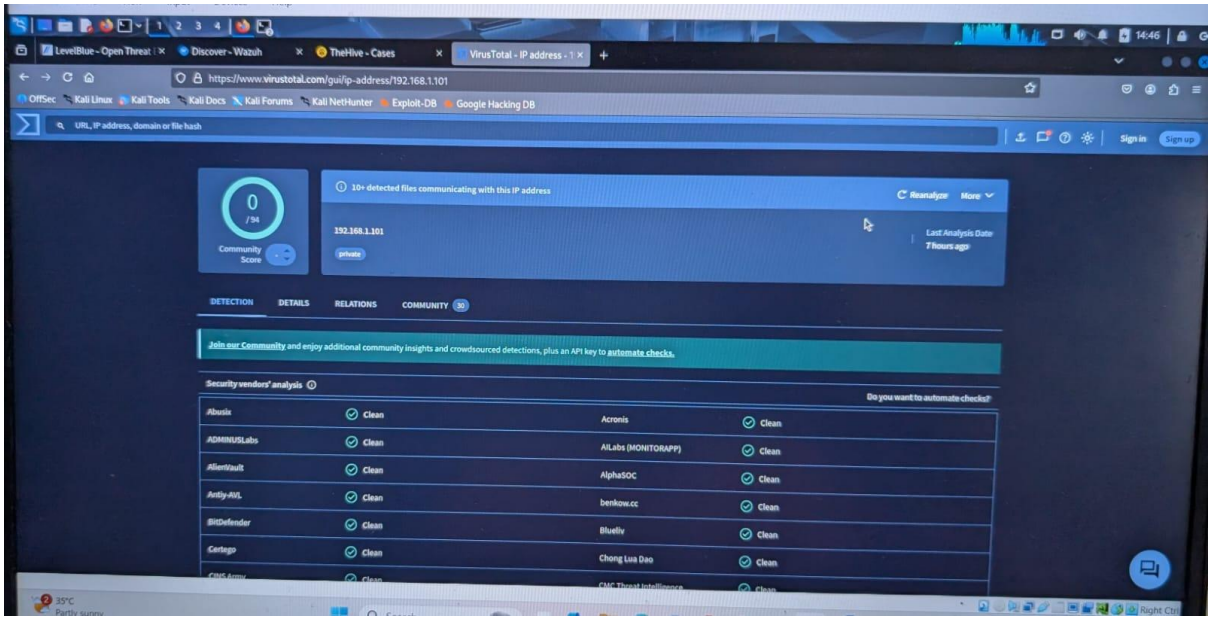
10.0.2.15:9000/index.html#!/case/~-20512/observables
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB
TheHive New Case My tasks Waiting tasks Alerts Dashboards Search
Case # 1 - [Critical] Ransomware Detected on Server-X
SOC Analyst 03/16/26 13:24 a minute
Details Tasks Observables TTPs
Filters
List of observables (2 of 2)
Flags Type Value/Filename Dates
ip 192[.]168[.]1[.]50
filename crypto_locker[.]exe
Isolate affected server
Investigate network traffic from IP 192.168.1.50
Identify infected host
Analyze suspicious file crypto_locker.exe
ip: 192[.]168[.]1[.]50
description: IP address
filename: crypto_locker[.]exe


LevelBlue - Open Threat
Discover - Wazuh
TheHive - Cases
VirusTotal - IP address -
https://www.virustotal.com/gui/ip-address/192.168.1.101
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
URL, IP address, domain or file hash
Sign in
Sign up
0
/94
Community Score
10+ detected files communicating with this IP address
Reanalyze
More
192.168.1.101
private
Last Analysis Date
7 hours ago
DETECTION
DETAILS
RELATIONS
COMMUNITY 30
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.
Security vendors' analysis
Do you want to automate checks?
Abusix
Clean
Acronis
Clean
ADMINUSLabs
Clean
AILabs (MONITORAPP)
Clean
AlienVault
Clean
AlphaSOC
Clean
Antiy-AVL
Clean
benkow.cc
Clean
BitDefender
Clean
Blueliv
Clean
Certego
Clean
Chong Lua Dao
Clean